

Final Model

Final Model Development and Deployment

After evaluating multiple feature selection and modeling approaches, the final model was selected based on the highest predictive performance.

Final Selected Approach

Feature Selection : RFE

Algorithm : Random Forest Regressor

Hyperparameter : GridSearchCV

Model Serialization

The trained pipeline was saved using **Pickle** for future prediction and deployment.

Model file:

finalized_model_RFE_Residential_House_Price_data.sav

Deployment Workflow

The deployment notebook performs the following operations:

1. Load saved model
2. Load cleaned dataset
3. Apply identical preprocessing
4. Select sample input
5. Generate prediction

Sample deployment code:

```
loaded_model.predict(sample)
```

Prediction Example

The deployment model successfully generated predicted residential house prices using selected property attributes.

Conclusion

The final Residential House Price Prediction system was developed using feature engineering, feature selection, and hyperparameter optimization techniques. The Random Forest model achieved reliable prediction performance and can be reused for future house price estimation tasks.
